

Resmetirom in NASH with Liver Fibrosis

Critical Appraisal of the MAESTRO-NASH Phase 3 Trial

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A Phase 3, Randomized, Controlled Trial of Resmetirom in NASH with Liver Fibrosis

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ABSTRACT

Trial Identity and Citation

TRIAL NAME & CITATION

MAESTRO-NASH Trial: A Phase 3, Randomized, Controlled Trial of Resmetirom in NASH with Liver Fibrosis

Citation: N Engl J Med 2024;390:497-509

FUNDING & RESEARCH GROUPS

- Full Citation: Harrison SA, Bedossa P, Guy CD, et al. A Phase 3, Randomized, Controlled Trial of Resmetirom in NASH with Liver Fibrosis. N Engl J Med 2024;390:497-509. doi:10.1056/NEJMoa2309000
- Trial Name: MAESTRO-NASH (Phase 3 Trial of Resmetirom in NASH with Liver Fibrosis).
- Trial Registration: ClinicalTrials.gov number, NCT03900429.

TRIAL REGISTRATION & SPONSORS

- Funding: Funded by Madrigal Pharmaceuticals (sponsor of the trial).
- Sponsors: Madrigal Pharmaceuticals. The sponsor designed the trial and performed data collection/analysis.
- Oversight: Approved by IRB/ethics committees at each participating site. Conducted in accordance with Declaration of Helsinki.

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INTRODUCTION

Background, Clinical Need, and Trial Rationale

Why This Paper Matters

Clinical problem: Clinical Problem: NASH with liver fibrosis is a major cause of cirrhosis, liver failure, and liver cancer. Prior to this trial, there was no approved pharmacotherapy.

Evidence gap: Evidence Gap: While lifestyle interventions are effective, long-term adherence is low, and no medical therapies have shown concurrent NASH resolution and fibrosis reduction.

Research question: Research Question: Does resmetirom (80 mg or 100 mg once daily), a selective thyroid hormone receptor beta agonist, resolve NASH and improve liver fibrosis?

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ABSTRACT

Clinical Question & PICO Framework

- P** — Adults with biopsy-confirmed nonalcoholic steatohepatitis (NASH) with a NAFLD Activity Score ≥ 4 and liver fibrosis stages F1B, F2, or F3.
- I** — Resmetirom at once-daily oral doses of 80 mg or 100 mg.
- C** — Placebo once daily.
- O** — Primary endpoints at week 52: (1) NASH resolution with no worsening of fibrosis, and (2) Fibrosis improvement by ≥ 1 stage with no worsening of NAS.

Introduction Summary

- NASH is characterized by fat accumulation, inflammation, and ballooning (cell damage), leading to progressive liver fibrosis.
- Fibrosis stages F2 and F3 (clinically significant fibrosis) represent a key threshold where liver-related morbidity and mortality rise exponentially.
- Thyroid hormone receptor beta (THR- β) is the dominant thyroid hormone receptor in the liver; its activity is crucial for mitochondrial health and lipid metabolism, but is down-regulated in NASH.
- Resmetirom is a highly selective oral THR- β agonist designed to restore hepatic thyroid hormone signaling, reduce hepatic fat accumulation, and halt the fibrotic cascade.

Introduction Critical Appraisal

- Background Justification: The clinical rationale is solid; targeting THR- β directly addresses the underlying pathophysiological metabolic dysfunction of NASH.
- Surrogate Endpoints: Using histologic endpoints (NASH resolution, fibrosis reduction) is regulatory-compliant (FDA path) but clinically surrogate; hard clinical outcomes (cirrhosis, transplant, death) are not yet reported in this 52-week analysis.
- Trial Duration: 52 weeks is standard for histologic changes, but NASH is a chronic disease requiring years to show hard outcome benefits (the trial continues to 54 months for these outcomes).

METHODS

Study Population, Eligibility, Outcomes, and Statistics

Methods Overview

- **Steering & Design:** Designed by the sponsor in collaboration with an academic steering committee. Independent event adjudication committees reviewed safety.
- **Ethics & Consent:** Approved by institutional review boards at all 245 participating centers. Written informed consent obtained from all subjects.
- **Pathology Consensus:** Screening and week 52 biopsies were reread independently by two expert pathologists. Disagreements went to a blinded consensus panel.

Eligibility Criteria

KEY INCLUSION CRITERIA

- Age ≥ 18 years with at least three metabolic risk factors (metabolic syndrome criteria).
- Non-invasive screening showing transient elastography (FibroScan) LSM ≥ 8.5 kPa and CAP ≥ 280 dB/m.
- Biopsy-confirmed NASH (NAS ≥ 4 , with score ≥ 1 for steatosis, lobular inflammation, and ballooning) and fibrosis stage F1B, F2, or F3.

KEY EXCLUSION CRITERIA

- Significant alcohol consumption (>20 g/day for women, >30 g/day for men).
- Glycated hemoglobin (HbA1c) level of more than 9.0% at screening.
- Other causes of chronic liver disease (e.g., viral hepatitis, autoimmune hepatitis, hemochromatosis, Wilson's disease).

Key Sub-populations: *The trial targeted patients with significant fibrosis; at least 50% of the enrolled cohort was required to have stage F3 fibrosis.*

Study Design & Methodology

TRIAL ARCHITECTURE

- Phase 3, international, multicenter, double-blind, randomized, placebo-controlled trial.
- Randomization was performed in a 1:1:1 ratio using an interactive web-based system.
- Stratification was based on status with respect to type 2 diabetes (presence/absence) and baseline fibrosis stage (F1B, F2, F3).

BIAS CONTROLS

- Double-blind design: Patients, investigators, and site personnel were blinded to trial-group assignments.
- Unblinding protections: Lipid and thyroid hormone levels, which could reveal treatment, were kept blinded to local investigators.
- Central Pathology: Biopsies were read independently by two central pathologists who were unaware of group assignment, sequence of biopsy, or each other's reads.

Trial Endpoints & Definitions

PRIMARY HISTOLOGIC ENDPOINTS

- NASH Resolution: NAS score reduction of ≥ 2 points, with ballooning = 0, lobular inflammation = 0-1, and no worsening of fibrosis.
- Fibrosis Improvement: Reduction in fibrosis stage by ≥ 1 stage (NASH CRN criteria) with no worsening of the NAFLD activity score.

KEY SECONDARY EFFICACY

- Percent change from baseline in LDL cholesterol at week 24.
- Lipid markers: ApoB, Triglycerides, and Lipoprotein(a) at week 24.
- Non-invasive tests: changes in hepatic fat (MRI-PDFF) and liver stiffness (VCTE/MRE) at week 52.

SAFETY ENDPOINTS

- Systematic collection of adverse events (AEs) and serious adverse events (SAEs).
- Discontinuations due to adverse events.
- Adjudicated safety events (deaths, CV events, drug-induced liver injury).

Analysis Sets & Missing Data

- Primary Analysis Population: 966 randomized patients who had a baseline fibrosis stage of F1B, F2, or F3 (efficacy and safety primary cohort).
- Primary Biopsy Analysis Population: 955 patients. 11 patients were excluded from biopsy analysis due to delayed week 52 biopsy (related to Covid-19 site closures).
- Missing Biopsy Handling: Non-Responder Imputation (NRI) - patients who missed their week 52 biopsy or withdrew from the study were counted as non-responders for the primary histologic endpoints.
- Sensitivity Analyses: Multiple imputation and tipping-point analyses were performed to confirm that the missing data did not bias the primary efficacy conclusions.

Statistical Methods & Multiplicity

- Power calculation: A sample size of ≥ 780 patients provided $>90\%$ power to detect a difference between each resmetirom dose and placebo.
- Primary Model: Cochran-Mantel-Haenszel (CMH) test, stratified by diabetes status and baseline fibrosis stage, was used for primary analysis.
- Multiplicity Control: Controlled using a weighted Bonferroni approach and a two-stage gatekeeping procedure to adjust for two doses and two primary endpoints.
- Alpha allocation: Efficacy significance threshold set at $\alpha = 0.04$ for the week 52 biopsy endpoints, preserving $\alpha = 0.01$ for the final 54-month analysis.

Methods Critical Appraisal

- Strengths: Blinding was extremely rigorous, with additional safeguards (hiding lipids/thyroid parameters) to prevent inadvertent unblinding.
- Strengths: Pathologist blinding was robust, including large unpaired slide groups spiked with screening failures and blinded to sequence (baseline vs. week 52).
- Limitations: Primary analysis population (966 patients) excluded the 84 F1A/F1C patients, who represent a very early-stage disease with lower clinical risk.
- Limitations: Biopsy reading is subjective; even with independent central pathologists, inter-observer variability is a well-known issue (mitigated by the consensus read).

Participant Flow & Retention

- Randomization: 1050 patients randomized: 966 in the primary efficacy population (322 in 80 mg, 323 in 100 mg, 321 in placebo).
- Biopsy cohort: 955 patients had week 52 biopsy evaluation (316 in 80 mg, 321 in 100 mg, 318 in placebo).
- Delayed Biopsies: 11 patients had delayed week 52 biopsies due to Covid-19 and were removed from primary biopsy analyses (considered missing completely at random).
- Adherence: Trial adherence was high, with 92% of patients achieving >80% adherence to the daily dosing schedule.
- Discontinuations: Study discontinuation rates due to any reason before week 52 were balanced, but drug discontinuation due to adverse events was higher in the 100 mg group (7.7%) compared to 80 mg (2.8%) and placebo (3.4%).

Table 1. Demographics & Clinical Characteristics (Part 1)

Baseline patient characteristics in the primary analysis population

Table 1. Demographic and Clinical Characteristics of the Patients at Baseline (Primary Population).*			
Characteristic	Resmetirom, 80 mg (N=322)	Resmetirom, 100 mg (N=323)	Placebo (N=321)
Age — yr	55.9±11.5	57.0±10.8	57.1±10.5
Male sex — no. (%)†	140 (43.5)	141 (43.7)	143 (44.5)
Race or ethnic group — no. (%)†			
White	291 (90.4)	291 (90.1)	281 (87.5)
Black	5 (1.6)	5 (1.5)	9 (2.8)
Asian	10 (3.1)	9 (2.8)	9 (2.8)
Other‡	12 (3.7)	11 (3.4)	18 (5.6)
Missing data	4 (1.2)	7 (2.2)	4 (1.2)
Hispanic or Latino ethnic group — no. (%)†	71 (22.0)	81 (25.1)	52 (16.2)
Body weight — kg	100.1±22.3	101.9±22.9	100.2±23.1
Body-mass index	35.5±6.4	36.2±7.4	35.3±6.5
Type 2 diabetes — no. (%)	224 (69.6)	213 (65.9)	210 (65.4)
Hypertension — no. (%)	243 (75.5)	254 (78.6)	257 (80.1)
Dyslipidemia — no. (%)	229 (71.1)	236 (73.1)	224 (69.8)
Hypothyroidism — no. (%)§	39 (12.1)	46 (14.2)	45 (14.0)
History of ASCVD — no. (%)	20 (6.2)	23 (7.1)	14 (4.4)
Estimated 10-yr risk of ASCVD — %¶	14.7±12.0	14.5±12.1	15.4±11.6
FibroScan liver-stiffness measurement — kPa			
Mean	13.3±6.8	13.6±7.1	12.9±5.5
Median (IQR)	11.5 (9.5–14.9)	11.9 (9.5–15.9)	11.7 (9.4–14.8)
FibroScan controlled attenuation parameter — dB/m**	346.1±37.2	349.4±38.7	347.2±37.0
MRI-PDFF — %††	18.2±6.8	17.2±6.6	17.8±6.8
Liver stiffness on MRE — kPa	3.5±0.9	3.7±1.1	3.5±1.0
Fibrosis-4 index score‡‡	1.4±0.7	1.5±0.7	1.4±0.7
LDL cholesterol level — mg/dl	106.6±37.4	103.0±36.8	106.8±41.1
Alanine aminotransferase level — U/liter	52.8±27.3	56.3±34.0	54.7±34.8
Aspartate aminotransferase level — U/liter	38.2±19.3	42.5±25.2	40.7±24.6
γ-Glutamyltransferase level — U/liter	84.3±111.3	84.6±99.0	75.7±85.0

Table 1 (Part 1). Demographics and clinical characteristics of patients at baseline.

Table 1. (Continued.) Baseline Liver Histology (Part 2)

Baseline histology and liver stiffness parameters

Table 1. (Continued.)			
Characteristic	Resmetirom, 80 mg (N = 322)	Resmetirom, 100 mg (N = 323)	Placebo (N = 321)
Liver-biopsy findings — no. (%)			
NAFLD activity score ≥ 5§§	266 (82.6)	288 (89.2)	253 (78.8)
Fibrosis stage¶¶			
F1B	16 (5.0)	15 (4.6)	18 (5.6)
F2	107 (33.2)	100 (31.0)	112 (34.9)
F3	199 (61.8)	208 (64.4)	191 (59.5)

* Plus-minus values are means ±SD. Percentages may not total 100 because of rounding. ASCVD denotes atherosclerotic cardiovascular disease, IQR interquartile range, LDL low-density lipoprotein, and MRE magnetic resonance elastography.

† Data on sex, race, and ethnic group were reported by the patient.

‡ “Other” includes American Indian or Alaska Native, Native Hawaiian or other Pacific Islander, and all other nonspecified race or ethnic group categories.

§ Shown are patients who were receiving thyroxine-replacement therapy at screening.

¶ The risk of cardiovascular events was derived from multiple risk factors, including age, sex, race, systolic blood pressure, diastolic blood pressure, total cholesterol, high-density lipoprotein cholesterol, LDL cholesterol, history of diabetes, smoking status, hypertension treatment, statin treatment, and aspirin therapy. Persons are preliminarily classified on the basis of estimated risk: a 10-year risk of ASCVD of less than 5% is low risk; 5 to 7.4% is borderline risk; 7.5 to 19.9% is intermediate risk; and 20% or more is high risk.²⁵

|| Liver stiffness was measured by means of vibration-controlled transient elastography. Values of more than 8.5 are considered to be indicative of fibrosis of stage F2 or higher.³

** Controlled attenuation parameter is a method for the noninvasive assessment of steatosis, which measures the increased attenuation of ultrasound waves when traveling through steatotic hepatic tissue, as compared with normal liver tissue. The maximum value is 360 dB per meter; for this trial, a reading of more than 280 dB per meter was considered to be high.³

†† Magnetic resonance imaging proton density fat fraction (MRI-PDFF) is a magnetic resonance imaging–derived noninvasive, quantitative biomarker to assess liver fat content. A reading of more than 5% is considered to be high.²⁶

‡‡ The Fibrosis-4 index score is derived from platelet count, aspartate aminotransferase level, age, and alanine aminotransferase level. Scores of more than 2.67 are considered to be indicative of advanced fibrosis and an elevated risk of liver-related events.³

§§ The nonalcoholic fatty liver disease (NAFLD) activity score is assessed on a scale of 0 to 8, with higher scores indicating more severe disease; the components of this measure are steatosis (assessed on a scale of 0 to 3), lobular inflammation (assessed on a scale of 0 to 3), and hepatocellular ballooning (assessed on a scale of 0 to 2). NAFLD activity scores of 4 or more are considered to indicate at-risk nonalcoholic steatohepatitis (NASH).²³

¶¶ Fibrosis stages range from F0 (no fibrosis) to F4 (cirrhosis). A stage of F1B indicates moderate fibrosis, pericentral area only. Five patients in each group who were scored as having F3 fibrosis at eligibility and who were rescored as having F4 fibrosis at baseline by one or both central pathologists were evaluated in the F3 group.

Table 1 (Part 2). Baseline liver biopsy findings and non-invasive liver parameters.

Baseline Comparability Appraisal

- Balance: Demographic and clinical characteristics were highly balanced across the three trial groups (age, BMI, sex, comorbidities).
- Diabetes Balance: T2DM was present in 69.6% (80 mg), 65.9% (100 mg), and 65.4% (placebo), which is a key stratification factor.
- Fibrosis Balance: F3 fibrosis (advanced pre-cirrhotic) was present in 61.8% (80 mg), 64.4% (100 mg), and 59.5% (placebo). This high proportion of F3 patients makes the cohort highly relevant for clinical progression.
- No Imbalances: No clinically meaningful imbalances in baseline parameters were observed that could confound the treatment effect.

Table 2. Week 52 Biopsy End Points

Primary and secondary histologic outcomes

Table 2. Biopsy End Points.*							
End Point	Resmetirom, 80 mg (N = 316)	Resmetirom, 100 mg (N = 321)	Placebo (N = 318)	Difference between Resmetirom, 80 mg, and Placebo (95% CI)†	P Value	Difference between Resmetirom, 100 mg, and Placebo (95% CI)†	P Value
	percent with response			percentage points		percentage points	
Primary end points							
NASH resolution with no worsening of fibrosis	25.9	29.9	9.7	16.4 (11.0–21.8)	<0.001	20.7 (15.3–26.2)	<0.001
Fibrosis improvement by ≥1 stage with no worsening of NAFLD activity score	24.2	25.9	14.2	10.2 (4.8–15.7)	<0.001	11.8 (6.4–17.2)	<0.001
Other end points							
≥2-Point improvement in NAFLD activity score, including ≥1-point improvement in hepatocellular ballooning or lobular inflammation, with no worsening of fibrosis	41.3	44.9	21.2	20.2 (13.8–26.5)		23.8 (17.4–30.2)	
≥2-Point improvement in NAFLD activity score, including ≥1-point improvement in hepatocellular ballooning or lobular inflammation, with improvement in fibrosis	18.8	21.2	8.5	10.5 (5.8–15.3)		13.0 (8.3–17.7)	
Improvement in each component of NAFLD activity score	23.3	27.9	7.2	16.1 (11.1–21.0)		20.9 (15.8–25.9)	
Improvement in fibrosis by ≥2 stages	8.3	10.1	2.8	5.6 (2.5–8.7)		7.4 (3.9–10.8)	
Both NASH resolution and fibrosis improvement by ≥1 stage	14.2	16.0	4.9	9.5 (5.4–13.6)		11.6 (7.5–15.8)	

* Of the 966 patients in the primary population, 11 patients (6 in the 80-mg resmetirom group, 3 in the 100-mg resmetirom group, and 2 in the placebo group) had a delay in their week 52 biopsy for reasons related to coronavirus disease 2019 (Covid-19) and were not evaluated for the end points shown here. NASH resolution was defined as a hepatocellular ballooning score of 0, a lobular inflammation score of 0 or 1, and a reduction in the NAFLD activity score by at least 2 points.

† The widths of the confidence intervals have not been adjusted for multiplicity and may not be used for hypothesis testing.

Table 2. Primary and secondary histologic biopsy endpoints at week 52.

Figure 1. Primary and Key Secondary End Points

Summary of NASH-resolution, fibrosis-improvement, and LDL cholesterol reduction

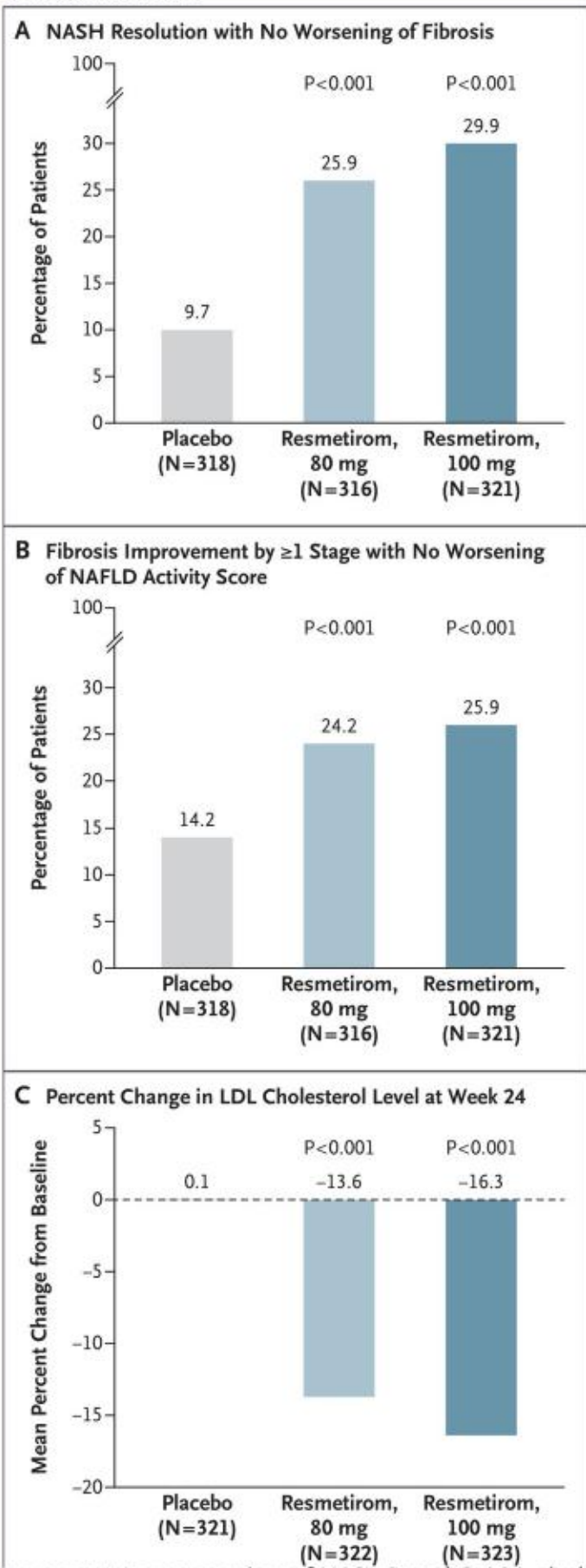


Figure 1. Primary and Key Secondary End Points. The two primary end points at week 52 were resolution of nonalcoholic steatohepatitis (NASH) with no worsening of fibrosis (Panel A), and an improvement (reduction) in fibrosis by at least one stage with no worsening of the nonalcoholic fatty liver disease (NAFLD) activity score (Panel B). The key secondary end point was the percent change from baseline in the low-density lipoprotein (LDL) cholesterol level at week 24 (Panel C). The NAFLD activity score is assessed on a scale of 0 to 8, with higher scores indicating more severe disease; the components of this measure are steatosis (assessed on a scale of 0 to 3), lobular inflammation (assessed on a scale of 0 to 3), and hepatocellular ballooning (assessed on a scale of 0 to 2). NASH resolution was defined as achievement of a hepatocellular ballooning score of 0, a lobular inflammation score of 0 or 1, and a reduction in the NAFLD activity score by at least 2 points. Fibrosis stages range from F0 (no fibrosis) to F4 (cirrhosis). A total of 11 patients had a delay in their week 52 biopsy due to coronavirus disease 2019–related closure of the biopsy site or related reasons and were removed from the primary analysis population for liver-biopsy analyses.

Figure 1. Panel A: NASH resolution (no worsening of fibrosis); Panel B: Fibrosis improvement (no worsening of NASH); Panel C: LDL cholesterol percent change at week 24.

Table 3. Lipids & Liver Enzymes at Weeks 24 & 48

Secondary lipid-lowering and biochemistry endpoints

Table 3. Key Secondary and Other Secondary End Points (Primary Population).*

Measurement	Resmetirom, 80 mg (N=322)	Resmetirom, 100 mg (N=323)	Placebo (N=321)	Difference between Resmetirom, 80 mg, and Placebo (95% CI)†	Difference between Resmetirom, 100 mg, and Placebo (95% CI)†
	<i>least-squares mean percent change from baseline</i>			<i>percentage points</i>	
LDL cholesterol level at wk 24‡§	−13.6±1.7	−16.3±1.7	0.1±1.7	−13.7 (−17.5 to −10.0)¶	−16.4 (−20.1 to −12.6)¶
Apolipoprotein B level at wk 24§	−16.8±1.3	−19.8±1.3	0.39±1.3	−17.2 (−20.0 to −14.4)	−20.2 (−22.9 to −17.4)
Triglyceride level at wk 24§	−22.7±4.0	−21.7±4.3	−2.6±4.1	−20.1 (−28.3 to −11.8)	−19.1 (−27.8 to −10.3)
Lipoprotein(a) level at wk 24§**	−30.4±3.8	−35.9±4.0	−0.84±3.5	−29.5 (−37.6 to −21.5)	−35.1 (−43.5 to −26.6)
MRI-PDFF at wk 52	−35.4±2.8	−46.6±2.8	−8.7±2.7	−26.7 (−32.9 to −20.6)	−37.9 (−44.2 to −31.7)
Alanine aminotransferase level at wk 48††	−26.6±3.7	−33.2±3.9	−6.9±3.8	−19.7 (−27.7 to −11.6)	−26.3 (−34.5 to −18.1)
Aspartate aminotransferase level at wk 48††	−22.1±3.9	−28.3±3.9	−2.9±3.8	−19.3 (−27.2 to −11.3)	−25.4 (−33.5 to −17.4)
γ-Glutamyltransferase level at wk 48††	−25.0±5.5	−31.9±6.3	3.3±5.2	−28.3 (−37.3 to −19.3)	−35.2 (−45.5 to −25.0)

* Multiple imputation analyses were used for lipids and liver enzymes. Details on the change from baseline in levels of lipids, lipoproteins, and lipid particles at weeks 24 and 52 are provided in Table S11.

† The widths of the confidence intervals have not been adjusted for multiplicity and may not be used for hypothesis testing.

‡ The key secondary end point was the percent change from baseline in the LDL cholesterol level at week 24. LDL cholesterol was directly measured.

§ Data were missing for one patient in the 80-mg resmetirom group.

¶ P<0.001.

|| Data are for patients with a baseline triglyceride level of more than 150 mg per deciliter.

** Data are for patients with a baseline lipoprotein(a) level of more than 10 nmol per liter.

†† Data are for patients with a baseline alanine aminotransferase level of 30 U per liter or more.

Table 4. Safety Summary & Adverse Events

Safety profile and most frequent adverse events

Table 4. Safety Summary (Primary Population).			
Event	Resmetirom, 80 mg (N = 322)	Resmetirom, 100 mg (N = 323)	Placebo (N = 321)
number of patients (percent)			
≥1 Adverse event	296 (91.9)	296 (91.6)	298 (92.8)
Grade 1: mild	73 (22.7)	66 (20.4)	77 (24.0)
Grade 2: moderate	180 (55.9)	183 (56.7)	169 (52.6)
Grade 3 or higher: severe	43 (13.4)	47 (14.6)	52 (16.2)
≥1 Adverse event attributed to resmetirom or placebo*	124 (38.5)	134 (41.5)	88 (27.4)
≥1 Serious adverse event	35 (10.9)	41 (12.7)	37 (11.5)
≥1 Serious adverse event attributed to resmetirom or placebo*	2 (0.6)	0	1 (0.3)
Adverse event leading to trial discontinuation before wk 52†	6 (1.9)	22 (6.8)	7 (2.2)
Adverse event leading to trial discontinuation during entire treatment period†	9 (2.8)	25 (7.7)	11 (3.4)
Fatal adverse event	1 (0.3)	2 (0.6)	1 (0.3)
Major adverse cardiovascular event‡	1 (0.3)	1 (0.3)	1 (0.3)
Other cardiovascular event‡	0	1 (0.3)	3 (0.9)
Adverse events affecting >10% of patients in any group			
Diarrhea	87 (27.0)	108 (33.4)	50 (15.6)
Covid-19	69 (21.4)	54 (16.7)	66 (20.6)
Nausea	71 (22.0)	61 (18.9)	40 (12.5)
Arthralgia	48 (14.9)	35 (10.8)	40 (12.5)
Back pain	35 (10.9)	27 (8.4)	38 (11.8)
Urinary tract infection	33 (10.2)	27 (8.4)	27 (8.4)
Fatigue	33 (10.2)	26 (8.0)	28 (8.7)
Pruritus	26 (8.1)	37 (11.5)	22 (6.9)
Vomiting	28 (8.7)	35 (10.8)	17 (5.3)

* Shown are events that were considered by investigators to be related to resmetirom or placebo.
† Data are for events that emerged after the first dose of resmetirom or placebo and within 30 days after the last dose.
‡ Major adverse cardiovascular events were defined as nonfatal stroke, nonfatal myocardial infarction, and death from cardiovascular causes. All cardiovascular events were adjudicated.

Table 4. Serious adverse events, discontinuations, and common gastrointestinal events (diarrhea/nausea).

Efficacy Interpretation: Absolute vs. Relative Effects

5

NNT for NASH Resolution (100 mg)

9

NNT for Fibrosis Improvement (100 mg)

6

NNT for NASH Resolution (80 mg)

10

NNT for Fibrosis Improvement (80 mg)

- NASH Resolution (100 mg): ARR = 20.2 percentage points (29.9% vs 9.7%) -> NNT = 5 (95% CI, 4 to 7).
- Fibrosis Improvement (100 mg): ARR = 11.7 percentage points (25.9% vs 14.2%) -> NNT = 9 (95% CI, 6 to 16).
- NASH Resolution (80 mg): ARR = 16.2 percentage points (25.9% vs 9.7%) -> NNT = 6 (95% CI, 5 to 10).
- Fibrosis Improvement (80 mg): ARR = 10.0 percentage points (24.2% vs 14.2%) -> NNT = 10 (95% CI, 6 to 21).
- Clinical Benefit: Both doses show substantial, statistically significant, and clinically meaningful absolute benefits, with the 100 mg dose showing slightly higher efficacy.

Precision and Statistical Uncertainty

- Primary Confidence Intervals: All primary endpoints had 95% CIs that exclude 0, confirming statistical significance.
- NASH Resolution CI (100 mg vs. Placebo): 20.7% difference (95% CI, 15.3% to 26.2%). The narrow CI indicates high precision.
- Fibrosis Improvement CI (100 mg vs. Placebo): 11.8% difference (95% CI, 6.4% to 17.2%). High precision, confirming benefit.
- P-values: $P < 0.001$ for both primary comparisons against placebo for both doses, indicating highly robust statistical findings.

Subgroup Analyses & Consistency

- Consistency: Efficacy was consistent across pre-specified subgroups including age, sex, BMI, and type 2 diabetes status.
- Diabetes status: Efficacy of resmetirom in NASH resolution and fibrosis improvement was consistent in patients with and without type 2 diabetes.
- Baseline Fibrosis: Resmetirom showed benefit in both F2 and F3 subgroups, which is crucial as F3 represents the highest risk of progression.
- Non-invasive correlation: Efficacy was supported by consistent dose-dependent reductions in liver fat (MRI-PDFF) and liver stiffness (FibroScan LSM).

Results Critical Appraisal

- NRI conservative approach: The use of Non-Responder Imputation for missing biopsies is highly conservative and confirms that results are not driven by attrition bias.
- Histologic vs. Clinical: The histologic improvements are clear, but the ultimate clinical benefits (reduction in decompensation, transplant, or death) are not yet proven (awaited in the 54-month follow-up).
- Pathologist Disagreement: While central reads were independent, the paper does not report the exact rate of pathologist disagreement on the primary reads (though sensitivity analysis using consensus reads showed identical results).

Risk of Bias 2 (RoB 2) Summary

Domain 1: Bias arising from the randomisation process Centralized web-based randomization with stratification by diabetes status and baseline fibrosis. Baseline characteristics were highly balanced.	LOW RISK
Domain 2: Bias due to deviations from intended interventions Double-blind trial. Adherence was high (92% of patients >80%). Blinding safeguards for thyroid and lipid parameters were implemented to prevent unblinding.	LOW RISK
Domain 3: Bias due to missing outcome data Attrition was low. Non-responder imputation (NRI) was applied to all patients with missing week 52 biopsy data, a highly conservative approach.	LOW RISK
Domain 4: Bias in measurement of the outcome Biopsies read centrally by two independent, blinded pathologists. Disagreements resolved by a third blinded consensus read. Blinded adjudication for safety.	LOW RISK
Domain 5: Bias in selection of the reported result The trial was prospectively registered (NCT03900429). The protocol and statistical analysis plan were pre-specified and followed.	LOW RISK
OVERALL RISK OF BIAS JUDGEMENT	LOW RISK OF BIAS

GRADE Certainty of Evidence

GRADE FINAL CERTAINTY

MODERATE CERTAINTY

We have moderate confidence that resmetirom improves NASH resolution and liver fibrosis. We downgraded by one level for indirectness because the primary endpoints are histologic surrogates rather than hard clinical outcomes (such as liver failure, transplant, or death), which are still being collected in the ongoing 54-month follow-up.

Histologic NASH resolution and fibrosis improvement at 52 weeks

Risk of Bias

Low risk of bias across all domains (randomization, blinding, missing data handling, and selection of results).

Inconsistency

No serious inconsistency. Results were highly consistent across the two primary endpoints and pre-specified subgroups.

Indirectness

Serious concern. Histologic biopsy findings at 52 weeks are surrogate markers. Clinical outcome data (cirrhosis, liver failure, death) are not yet available.

Imprecision

No serious imprecision. The sample size was large, event rates were high, and confidence intervals for primary endpoints were narrow.

Publication Bias

Undetected. The trial was prospectively registered, protocol-driven, and sponsored academically/commercially with publication of all planned outcomes.

DISCUSSION & BROADER CONTEXT

Prior Evidence, Guidelines, and Bayesian Reanalysis

Chronology of Key Trials in NASH/MASH

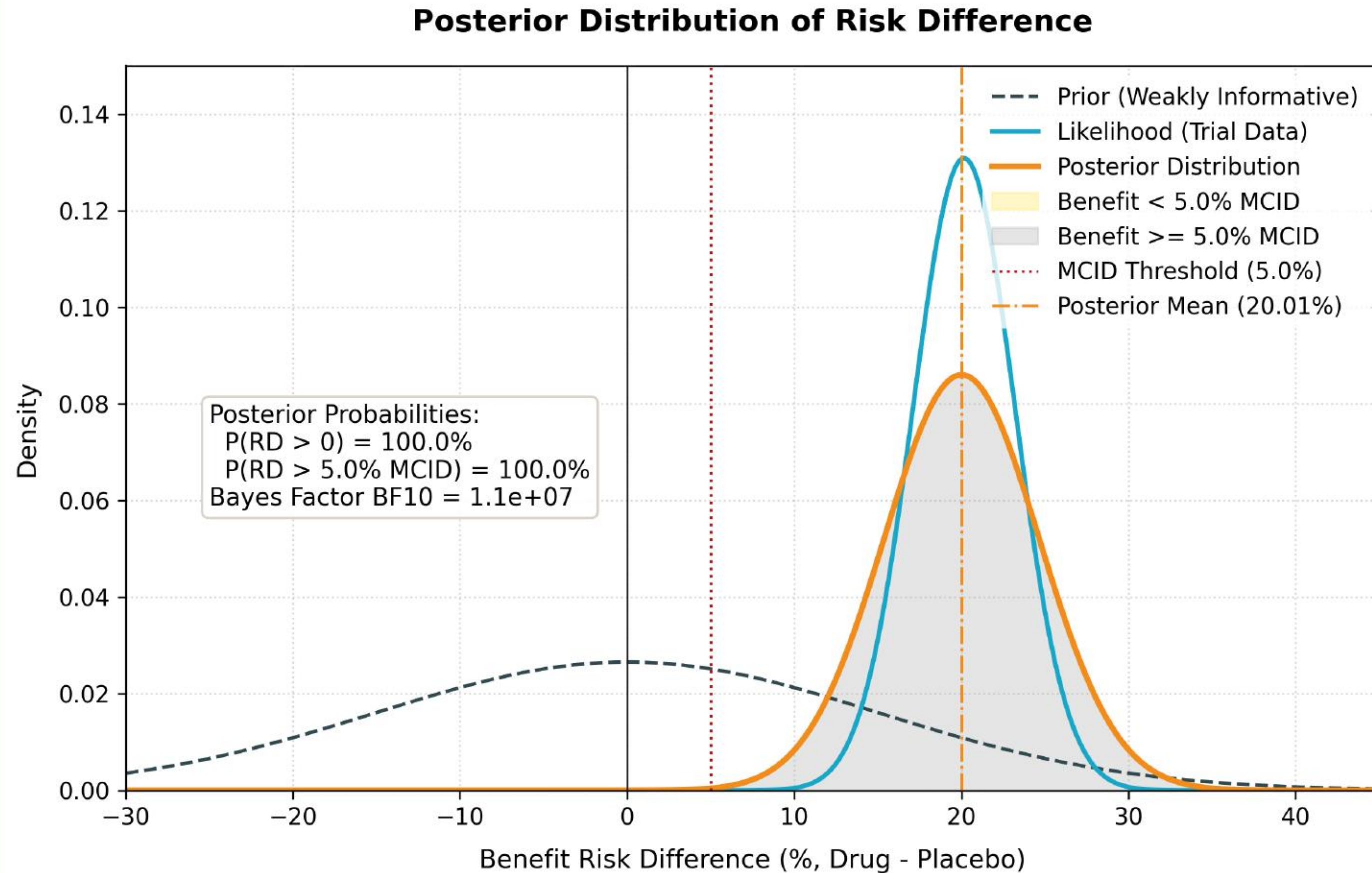
- 2010: PIVENS Trial (NEJM) - Pioglitazone and Vitamin E in non-diabetic NASH. Vitamin E improved NASH (43% vs 19% placebo, $P=0.001$) but did not improve fibrosis.
- 2015: FLINT Trial (Lancet) - Obeticholic acid (FXR agonist) in NASH. Improved liver histology (45% vs 21% placebo) but raised LDL cholesterol and caused pruritus.
- 2019: MAESTRO-NASH Phase 2 (Lancet) - Resmetirom significantly reduced hepatic fat fraction (MRI-PDFF) and resolved NASH (27.4% vs 6.1% placebo) at 36 weeks.
- 2023: REGENERATE Trial (Lancet) - Obeticholic acid in NASH. Improved fibrosis by ≥ 1 stage (22.4% with 25 mg vs 11.9% placebo) but failed to show NASH resolution. Pruritus and LDL elevation remained key concerns.
- 2024: MAESTRO-NASH Phase 3 (NEJM) - Current study. Landmark trial showing concurrent NASH resolution (29.9% vs 9.7%) and fibrosis improvement (25.9% vs 14.2%), leading to FDA approval.

Current Guideline Consensus & Regulatory Status

- **FDA Approval:** In March 2024, the FDA granted accelerated approval to resmetirom (Rezdiffra) for the treatment of adults with noncirrhotic NASH with moderate to advanced liver fibrosis (consistent with stages F2 to F3), based on this MAESTRO-NASH trial.
- **AASLD Guidance:** The American Association for the Study of Liver Diseases recommends considering resmetirom in patients with biopsy-confirmed NASH and significant fibrosis (F2-F3) to resolve NASH and improve fibrosis.
- **EASL Recommendations:** The European Association for the Study of the Liver similarly supports the use of resmetirom in this population, highlighting the need for non-invasive test (NIT) monitoring.
- **Indian Context:** No domestic biosimilar or low-cost generic is currently available in India, meaning cost remains a massive barrier for North Indian patients.

Posterior Distribution of Risk Difference (Weakly Informative Prior)

Conjugate prior/likelihood distribution on the risk difference parameter for a continuous endpoint (MCID = 5.0%)



Posterior Efficacy under Alternative Priors

Sensitivity analysis of posterior probabilities under different prior assumptions

Prior Profile	Prior Mean (SD) *	Posterior Mean RD (95% CrI)	P(RD > 0)	P(RD > 5.0% MCID)
Weakly Informative (Default)	0.00 (2.0000)	20.01% (11.54% to 29.65%)	100.0%	100.0%
Skeptical (MCID-derived)	0.00 (0.3675)	13.12% (6.99% to 20.21%)	100.0%	99.7%
Enthusiastic (MCID-derived)	0.47 (0.8982)	19.25% (11.12% to 28.52%)	100.0%	100.0%
Historical (Phase 2 Data)	0.80 (0.4000)	17.52% (10.43% to 25.58%)	100.0%	100.0%

* Note: Prior mean and SD are specified on the log-odds-ratio scale. Risk differences (RD) are drug minus placebo (positive values favor treatment). MCID is 5.0% absolute risk difference.

APPLICABILITY & CLINICAL PRACTICE

Local Context and Clinical Implementation

Local Applicability: North Indian / Ghaziabad Context

PRACTICAL CLINICAL BARRIERS

- Liver Biopsy Barrier: Liver biopsy is invasive, carries small risk, and is unacceptable to most patients in Ghaziabad for routine monitoring.
- Cost & Affordability: Resmetirom is a newly approved patent-protected drug with high monthly cost, out of reach for most patients at a North Indian medical college.
- Underrepresentation: Very few South Asian patients were included in the global MAESTRO-NASH trial (White 89%, Asian 3%).

FEASIBLE ADAPTATIONS

- Non-Invasive Screening: Utilize FibroScan transient elastography (CAP ≥ 280 dB/m for steatosis, LSM ≥ 8.5 kPa for fibrosis) for diagnosis instead of biopsy.
- Co-benefit on Lipids: Resmetirom's strong LDL-C and ApoB lowering effects are highly beneficial for North Indian metabolic syndrome patients who have very high cardiovascular risk.
- Supportive Care: Emphasize intensive lifestyle modification, diet counseling, and weight loss, which are the cornerstones of NASH management locally.

Indian Guideline Note: Non-invasive diagnostics (FibroScan LSM and CAP) must replace liver biopsy for screening and monitoring resmetirom therapy in Indian clinical practice.

Clinical Takeaways by Practice Setting

OUTPATIENT CLINIC (OPD)

- Screen high-risk metabolic patients (Type 2 diabetes, obesity) with FibroScan.
- Counsel patients starting resmetirom about initial mild diarrhea and nausea (median duration 15-20 days).
- Monitor lipid panel at 12 and 24 weeks to assess cardioprotective lipid-lowering benefit.

MEDICAL WARDS (IPD)

- Assess thyroid function parameters (FT4 reduction is expected, but TSH/FT3 remain normal; do not mistake for central hypothyroidism).
- Ensure patients admitted for other reasons continue daily resmetirom without interruption.
- Verify absence of decompensated cirrhosis (Child-Pugh B or C), which is a contraindication.

ICU / HDU SETTING

- Hold resmetirom in the setting of acute liver injury, decompensated liver disease, or severe sepsis.
- Monitor for drug-induced liver injury, although none was reported in this trial.
- Ensure coordination with gastroenterology/hepatology consults if liver function tests deteriorate acutely.

Journal Club Discussion Questions

Q1. Biopsy vs. Non-invasive Tests

In real-world practice, we cannot perform serial liver biopsies to monitor response. Can we safely rely on non-invasive markers like FibroScan LSM/CAP or MRI-PDFF to guide resmetirom initiation and cessation in our Ghaziabad patients?

Q2. Cost-Benefit in Resource-Limited Settings

Given the high cost of resmetirom, how should we prioritize it in our clinic? Should it be reserved strictly for patients with rapidly progressing F3 fibrosis who have failed intensive lifestyle modifications, or introduced earlier?

Q3. Long-term Cardiovascular Safety

Resmetirom showed significant improvements in LDL cholesterol, ApoB, and hepatic fat, but are these surrogate improvements guaranteed to translate into a reduction in major adverse cardiovascular events (MACE) and liver-related mortality at 54 months?

Q4. Child-Pugh B/C Contraindication

Since resmetirom is contraindicated in decompensated cirrhosis, how can we ensure strict vigilance in our outpatient screening to avoid accidental prescription to patients with subclinical portal hypertension or early Child-Pugh B cirrhosis?

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